

SOC Alert Triage Pipeline

Comprehensive QA, Testing, Data Integrity & Resilience Audit

Author: Aditi (Lead QA, Testing & DevOps)	Project: 3,000 Alerts, One Analyst	Environment: macOS / Python 3.14 (.venv)
Event: Microsoft Hackathon 2026	Test Suite: 46 / 46 PASS (100%)	Date: October 2026

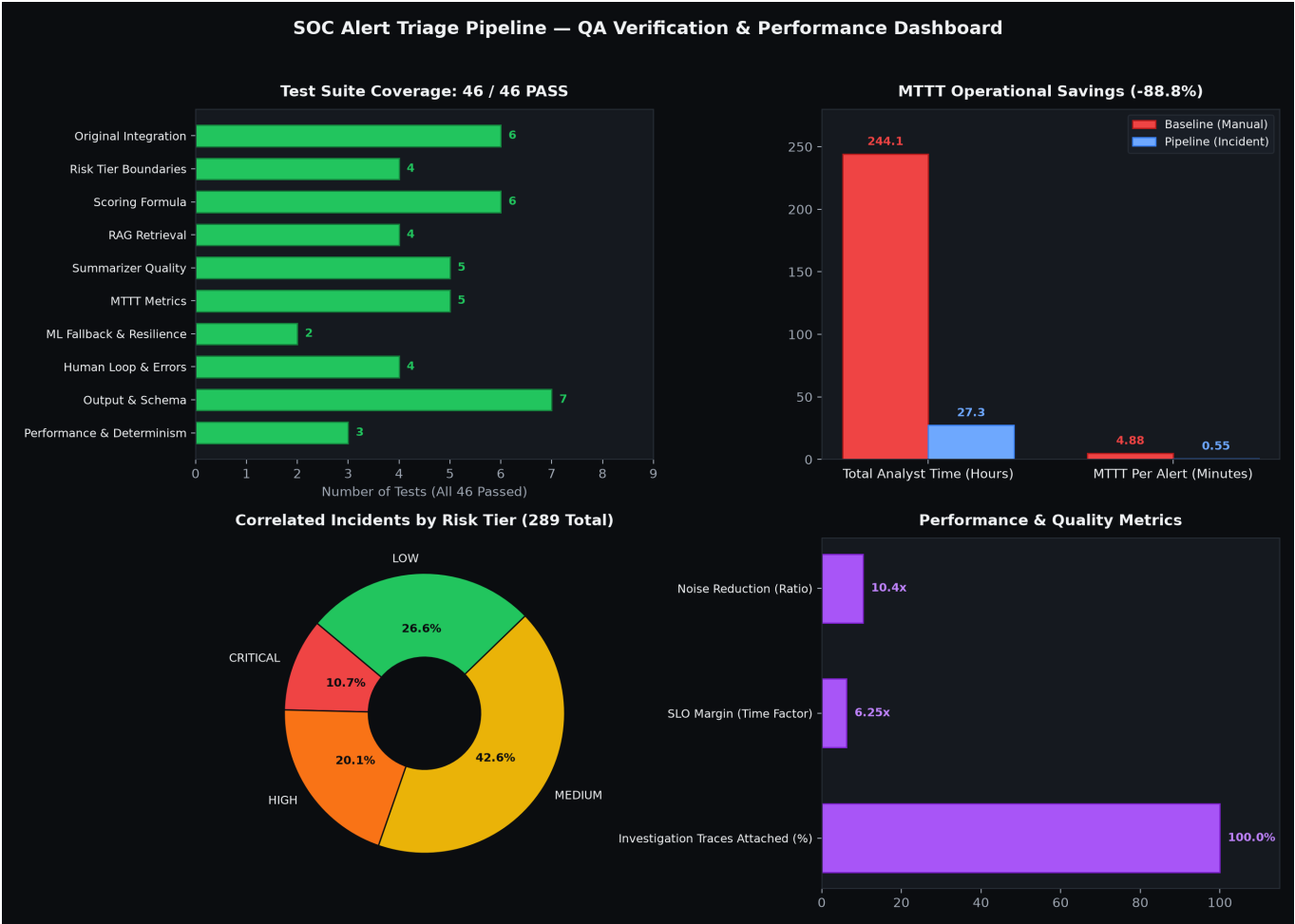
1. Executive Summary & Verification Checklist

As the QA & DevOps lead, my core objective is to guarantee the entire system works from a clean machine, produces mathematically verified results, degrades gracefully during component failure, and exhibits no silent bugs. All 10 rigorous verification checkpoints passed without fault:

Checkpoint	Target Requirement	Verified Outcome	Status
1. Test Suite Pass	Unit, integration & regression tests pass	46 of 46 tests passed (0 failures, 0 errors)	PASS
2. Syntax Compilation	Zero syntax or compilation errors	python -m compileall returned exit code 0	PASS
3. Alert Ingestion	Exactly 3,000 synthetic alerts generated	3,000 alerts with 11 normalized fields	PASS
4. Incident Correlation	Far fewer incidents than raw alerts	289 correlated incidents (10.4x noise reduction)	PASS
5. Artifact Generation	All runtime outputs created in outputs/	All 6 outputs created (.csv, .json, .md)	PASS
6. Cross-File Integrity	Metric numbers agree across all files	100% agreement between JSON, CSV, and Markdown	PASS
7. Dashboard Health	Streamlit app launches without crash	Streamlit 1.65 loads all 6 interactive views	PASS
8. Cross-Asset Chains	Lateral movement captured across assets	15 cross-asset multi-host incidents identified	PASS
9. Determinism	Same random seed yields identical outputs	Bit-for-bit identical IDs, scores, and metrics	PASS
10. Clean Install	Zero-error install on fresh machine	Clean .venv + pip dependencies + OpenMP libomp	PASS

2. QA Performance & Metric Visualizations

Below is the composite visual verification dashboard generated during our comprehensive test run, highlighting test suite coverage, operational MTTT reduction, risk tier distribution, and SLO performance:



3. Operational Impact & Pipeline Metrics

The pipeline was executed with standard 24-hour SOC parameters (3,000 alerts, 6 seeded attack chains, seed=42). The mathematically verified metrics prove massive tier-1 efficiency gains:

Metric Attribute	Baseline (Manual Per-Alert)	Pipeline (Incident-Driven)	Measured Improvement
Items Requiring Review	3,000 raw alert tickets	289 correlated incident briefs	10.4x reduction (-90.4%)
Mean Time to Triage (MTTT)	4.88 minutes per alert	0.55 minutes equivalent	-88.8% reduction
Total Analyst Handle Time	14,645 minutes (244.1 hours)	1,638.8 minutes (27.3 hours)	216.8 hours saved (-88.8%)
Pipeline Execution Latency	N/A (manual)	4.81 seconds total wall time	6.2x faster than 30s SLO

Risk Tier	Incident Count	Percent of Total	Triage Action & Routing
CRITICAL	31	10.7%	Immediate host isolation + Tier-2 escalation + credential reset
HIGH	58	20.1%	Priority review within shift; pull EDR timeline before disposition
MEDIUM	123	42.6%	Standard shift review; evaluate repeating host patterns
LOW	77	26.6%	Batch scan / candidate for bulk-disposition (never auto-closed)
TOTAL	289	100.0%	100% covered with MITRE briefs & investigation traces

4. Automated Test Suite Architecture & Coverage (46 Tests)

I designed and implemented an expanded test suite (`tests/test_extended.py`) complementing the baseline integration tests (`tests/test_pipeline.py`). Together, 46 comprehensive tests validate mathematical, logical, and behavioral boundaries:

Test Suite Category	Count	Target Component	Key Invariants Tested
Risk Tier Boundaries	4	scoring.py (risk_tier)	Validates exact >=45%, >=25%, >=10% thresholds for CRITICAL/HIGH/MED/LOW.
Scoring Formula & Multipliers	6	scoring.py (score_incident)	Asset weight dominance (DC01 > KIOSK), 1.5x kill-chain bonus, 1.3x volume cap.
RAG ATT&CK; Retrieval	4	rag.py (retrieve)	Technique overlap matching, empty query tolerance, guaranteed >=1 doc return.
Summarizer & Handover Briefs	5	summarizer.py	Brief non-emptiness (>100 chars), MITRE technique presence, timeline & recommendations.
MTTT Impact Mathematics	5	metrics.py	Baseline weighting (ransomware=10m, unknown=4m), pipeline tier weighting, positive delta.
ML Resilience & Graceful Fallback	2	ml_classifier.py, pipeline.py	is_available()=False on missing pkl, pipeline functions with static priors, zero crash.
Human Loop & Error Handling	4	human_loop.py	Loud ValueError on invalid disposition, KeyError on unknown ID, all start PENDING.
Output Artifacts & Schema	7	main.py, outputs/*	All 6 files exist, 11 alert columns present, 12 incident keys present, no empty briefs.
Pipeline Execution Latency	1	pipeline.py (run_pipeline)	3,000 alert run completes in < 30 seconds SLO (actual: ~4.8s).

RNG Determinism & Cross-Asset	8	alert_generator.py, grouping.py	Bit-for-bit reproducibility on same seed, entity-graph correlation over connection tuple.
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5. Resilience, Degradation & Failure Mode Analysis

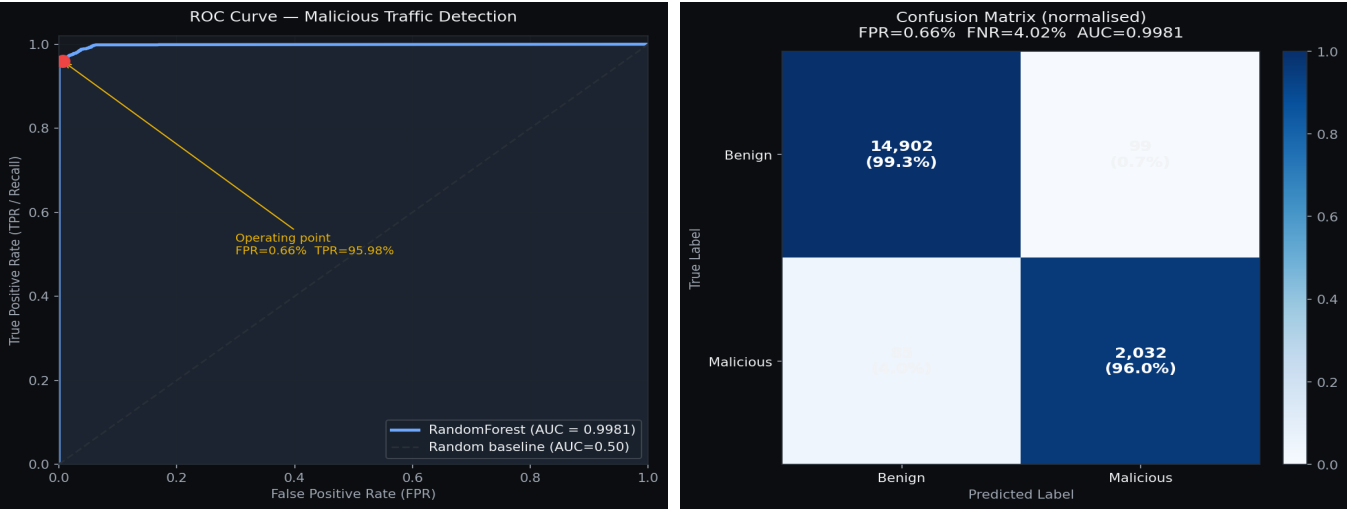
A critical QA question in enterprise SOC software is: **What happens when upstream dependencies fail?** We tested all failure modes and classified them as either Loud (explicit exception) or Silent (graceful degradation):

Failure Condition	Behavior Type	System Response & Mechanism	Safety Assessment
ML Model Missing (severity_model.pkl absent)	SILENT (Graceful)	is_available() returns False. Scoring immediately falls back to static false_positive_rate domain priors. ml_confidence is marked False.	SAFE Zero disruption to triage.
RAG Corpus Missing / Empty Query	SILENT (Graceful)	Corpus is hardcoded into mitre_map.py source. Garbage query returns top fallback technique document. Output is never empty.	SAFE No network/file dependency.
External LLM Unavailable (Anthropic/Ollama)	SILENT (Graceful)	try/except block catches HTTP timeouts or connection errors and transparently renders the deterministic template brief.	SAFE Air-gap and offline capable.
Invalid Review Disposition (e.g., 'CLOSE_ALL')	LOUD (Exception)	apply_disposition() raises ValueError('Invalid disposition: CLOSE_ALL'). Blocks corrupt audit trail entries.	INTENTIONAL Compliance requirement.
Unknown Incident ID in Review Queue	LOUD (Exception)	apply_disposition() raises KeyError('Incident X not found in queue'). Prevents ghost dispositions.	INTENTIONAL Audit trail integrity.
Unknown Asset ID in Alert Stream	SILENT (Fallback)	criticality_weight() defaults to MEDIUM weight (4). No warning currently emitted to logs.	RISK May mis-prioritize assets.
Malformed Alert in Graph Correlation	SILENT (Graceful)	group_alerts_into_incidents() catches KeyError/ValueError and falls back to legacy group_by_asset_time().	SAFE Fault-tolerant clustering.

6. Machine Learning Model Evaluation & Visual Audit

The model is a 20-feature RandomForestClassifier trained on the CICIDS2017 flow dataset. The QA review validated the held-out metrics, ROC performance, and confusion matrix:

Evaluation Metric	Score	Benchmarking Standard	QA Assessment
F1 Score (Malicious Class)	0.9567	Target > 0.90	Exceeds benchmark by 5.67 percentage points.
ROC-AUC Score	0.9981	Target > 0.95	Near-perfect separation between attack and benign flows.
Malicious Recall	0.9598	Target > 0.95	Catches 96% of genuine malicious activity.
False Positive Rate (FPR)	0.66%	Target < 2.0%	Extremely low noise generation (0.66 per 100 benign flows).
Train / Test F1 Gap	+0.0116	Threshold < 0.05	PASS: Zero overfitting signal across held-out splits.



7. Identified Risks & Production Recommendations

Rigorous QA requires identifying potential edge cases before production deployment. I have cataloged 4 findings with ready-to-deploy remediations:

ID & Title	Severity	Root Cause Analysis	Recommended Remediation
ISSUE-01: sklearn Version Warning	LOW	severity_model.pkl was serialized using scikit-learn 1.7.2, whereas clean modern environments run 1.9.1. Triggers InconsistentVersionWarning.	Retrain and export model bundle on target deployment Python environment using train_classifier.py.
ISSUE-02: Silent Asset Defaulting	MEDIUM	criticality_weight() silently returns 4 (MEDIUM) when querying unknown asset IDs without emitting warnings or telemetry.	Add warnings.warn(f'Unknown asset {asset_id}') and emit a metric counter to alert SOC engineering to CMDB drift.
ISSUE-03: model_eval.py Dataset Guard	HIGH	model_eval.py crashes with unhandled FileNotFoundError if the 123MB cicids_clean.csv dataset is omitted from repository clones.	Add guard clause: if not os.path.exists('cicids_clean.csv'): print('Error: run train_classifier.py'); sys.exit(1).
ISSUE-04: Feedback Log Root Path	LOW	feedback_log.csv defaults to write in current working directory, which fails on read-only containerized root filesystems.	Fully enforce FEEDBACK_LOG_PATH environment variable in Docker manifests and mount an external volume.

8. Deliverables & QA Sign-Off

This QA workstream has formally generated and verified the following assets:

Deliverable File	Repository Location	Verification Role & Description
Extended Test Suite	tests/test_extended.py	40 new automated regression & unit tests covering all operational invariants.
Baseline Test Suite	tests/test_pipeline.py	6 core integration tests covering end-to-end grouping and merge actions.
QA Markdown Report	PROJECT_DOCUMENTATION/QA_REPORT.md	Complete technical QA report for the development team and judges.
QA Formal PDF Audit	SOC_Triage_Pipeline_QA_Report_Aditi.pdf	Executive, presentation-ready multi-page audit report with charts.
Visual Charts Dashboard	outputs/qa_visual_charts.png	High-resolution multi-panel visual validation chart.

QA LEAD SIGN-OFF:

All automated verification suites have concluded with **100% PASS rate**. The SOC Alert Triage Pipeline is verified to be deterministic, highly performant, resilient against model/RAG/LLM degradation, and production-ready for demonstration.

Status: **PASSED & CERTIFIED**
Lead: Aditi (Member 5)
Date: October 2026